NOVA IMS Information Management School

MDSAA

Master Degree Program in

Data Science and Advanced Analytics

Beyond Static Investing: Forecasting Market Regimes for Adaptive Portfolios

Combining GMM Clustering and Markov Chains in Tactical Asset Allocation across a Diversified Asset Universe

Andriani Kakoulli

Master Thesis

presented as partial requirement for obtaining a Master's Degree in Data Science and Advanced Analytics

NOVA Information Management School
Instituto Superior de Estatística e Gestão de Informação

Universidade Nova de Lisboa

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STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism or any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

Lisbon, June 23rd, 2025

Andriani Kakoulli

DEDICATION

To my beloved parents, for your unwavering faith in me, your endless love, and the peace you bring to my life. Your confidence in my dreams and your joy in my achievements have made every step of this journey worthwhile. This is as much yours as it is mine.

Forever thankful,

Your little one,

Andriani

ABSTRACT

The main purpose of this master thesis is to develop a data-driven framework for identifying and forecasting market regimes using statistical and machine learning methods. The study analyzes a diversified set of financial assets, including U.S. large, mid, and small cap equity indices, international equity indices, and major cryptocurrencies, over the period from 2016 to 2024. The methodology consists of three main steps: Principal Component Analysis is first applied, and PCA loadings are examined to select the most informative features; a Gaussian Mixture Model is then used to cluster the transformed data and identify distinct market regimes; finally, a first-order Markov chain is constructed to estimate the transition probabilities between regimes, enabling regime forecasting. Based on the identified regimes, a Tactical Asset Allocation strategy is implemented, where portfolio weights are optimized for each regime and dynamically adjusted according to predicted regime probabilities. The main takeaway from this research is that a probabilistic regime-switching framework can systematically improve the flexibility and risk management of a multi-asset portfolio beyond a simple static allocation.

KEYWORDS

PCA loadings; Clustering; Gaussian Mixture Model; Markov Chain; Transition Matrix; Tactical Asset Allocation;

Sustainable Development Goals (SDG):

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LIST OF ABBREVIATIONS AND ACRONYMS

AIC	Akaike information criterion
BIC	Bayesian Information Criterion
BTC	Bitcoin cryptocurrency
SP500	S&P 500 index
EEM	MSCI Emerging Markets Index
EFV	iShares MSCI EAFE (Europe, Australasia and Far East) value ETF
ETH	Ethereum cryptocurrency
GMM	Gaussian Mixture Model
IJJ	iShares Morningstar Mid-Cap ETF
IMCV	iShares S&P Mid-Cap Value ETF
IWM	Russel 2000 index
IYJ	iShares U.S. Industrials ETF
MS-VARs	AnalysisMarkov-switching Vector Autoregressions
PCA	Principal Component Analysis
TAA	Tactical Asset Allocation
VV	Vanguard Large-Cap Index Fund ETF Shares

1. INTRODUCTION

Financial markets do not move in a smooth, continuous manner. Instead, they evolve through regimes, periods of relatively stable statistical behaviour that can shift abruptly or gradually in response to changes in investor sentiment, liquidity, macroeconomic conditions, or structural shocks. These regime shifts leave observable traces in patterns of returns, volatility, and cross-asset correlations (Ang & Bekaert, 2002; Guidolin & Timmermann, 2007). For investors and especially portfolio managers, recognizing and anticipating these shifts is crucial: the difference between capturing upside in risk-on periods and avoiding costly drawdowns in crisis phases often lies in how well one can detect and adapt to changing regimes.

Recent events such as the COVID-19 pandemic crash, the extraordinary 2020-2021 recovery, and the 2022-2023 inflation-driven drawdowns illustrate the reality that static "one-size-fits-all" portfolio strategies are increasingly unsuited to modern markets. Investors who stayed fully exposed during abrupt regime shifts suffered severe losses, while those who used adaptive approaches such as systematically reducing equity exposure during volatility spikes or tactically reallocating into safer assets weathered these transitions more effectively.

This thesis addresses a central challenge faced by investors and portfolio managers:

- Can financial market regimes be identified and forecasted in a systematic, data-driven way that improves timely portfolio adjustments and decision-making?

Specifically, it seeks to determine whether distinct, interpretable regimes can be inferred from the joint behaviour of equity indices of various capitalizations and major cryptocurrencies, and whether understanding the dynamics between these regimes can guide more adaptive, responsive investment strategies.

A key motivation for this study is that, while traditional parametric models such as Markov-switching models (Hamilton, 1989) and threshold-based indicators have been foundational in financial econometrics, they often rely on strict assumptions that do not reflect the noisy, high-dimensional nature of mixed-asset markets today (Kritzman et al., 2012). More recent studies have turned to unsupervised learning techniques and high-frequency clustering to extract latent structure directly from the data without imposing strong assumptions (Horvath, Issa, & Muguruza, 2021). Few studies have explored the simultaneous behaviour of traditional financial indices and digital assets, adding an additional layer of complexity and opportunity that remains underexplored in regime-aware frameworks.

Practical examples of adaptive strategies include tilting portfolios towards high-quality or defensive equities when probability of a risk-off regime rises, or allocating more capital to growth-oriented or higher-beta segments during sustained bull phases. Some hedge funds and risk parity funds already exploit these dynamics by systematically adjusting risk exposure and leverage based on changing volatility and cross-asset signals. Such strategies stand in

contrast to static allocation, which holds fixed weights regardless of shifts in market structure and therefore may fail to protect capital when structural breaks occur.

This thesis contributes to the existing literature in three ways. First, it introduces a comprehensive cross-asset analysis that spans large-cap, mid-cap, and small-cap U.S. equity indices alongside major cryptocurrencies, enabling a broader perspective on market structure. Second, it applies an unsupervised clustering method based on return and risk characteristics and models regime dynamics using a first-order Markov chain to estimate how likely the market is to persist in or shift between states, providing a probabilistic, forward-looking forecast. Third, it links these regimes directly to tactical asset allocation by constructing a regime-weighted allocation framework that reflects not just the current market environment, but also where the market is probabilistically heading. By applying the described steps, this study provides a novel blend of behavioural insight and predictive utility for real-world portfolio management (Faber, 2007).

The remainder of this thesis is structured as follows:
The Literature Review outlines theoretical and empirical foundations of regime detection, regime switching models and their relevance to asset allocation. The Methodology section details the dataset, preprocessing steps, feature engineering, the framework for regime identification, regime transition forecasting, and tactical allocation procedure. The Results and Discussion section presents the identified regimes, transition dynamics, and back-tested performance of the regime-weighted allocation. Finally, the Conclusion and Future Work section discuss key takeaways and suggest directions for extending this research, including integration of macroeconomic factors or more sophisticated regime-switching models for further robustness.

2. LITERATURE REVIEW

Financial market behavior is widely understood to exhibit distinct regimes or states, characterized by changing patterns in return distributions, volatility, and cross-asset relationships that cannot be captured by a single stable statistical process (Hamilton, 1989; Ang & Bekaert, 2002). This foundational insight that markets do not follow a single, stable statistical process, has led to a diverse body of research on how to identify, model, and respond to regime changes, recognizing that different market conditions often require different investment responses. While a variety of econometric and machine learning approaches have emerged to detect such states, questions remain about how best to apply these methods to complex, mixed-asset portfolios that include both traditional equity segments and newer asset classes such as cryptocurrencies.

2.1. REGIME-SWITCHING FOUNDATIONS

Seminal work by (Hamilton, 1989) introduced the Markov-switching autoregressive model to capture economic expansions and recessions, demonstrating that the probability of transitioning between states can be effectively modeled as a first-order Markov chain. Since then, regime-switching models have been applied to asset pricing, risk management, and dynamic portfolio strategies, illustrating that financial returns often exhibit state-dependent means, volatilities, and correlations that single-regime models cannot fully capture (Ang & Bekaert, 2002; Guidolin & Timmermann, 2007). Although these parametric models remain powerful, traditional parametric regime-switching approaches rely on strict distributional assumptions and can become inflexible or computationally intensive when extended to high-dimensional, noisy market data (Kritzman et al., 2012). This motivates the exploration of more flexible, data-driven alternatives that retain the interpretability and tractability of Markov models but relax the strict parametric constraints.

2.2. UNSUPERVISED REGIME DETECTION AND CLUSTERING

To overcome the constraints of rigid parametric models, more recent research has explored unsupervised learning methods to uncover regimes directly from the data structure (Horvath et al., 2021). Clustering plays a central role in this context because it provides a data-driven way to group periods of similar market behaviour without relying on subjective thresholds or predefined labels. By analysing patterns in risk and return features, clustering algorithms can partition historical observations into distinct states that share common statistical characteristics meaningful to portfolio managers (Kritzman et al., 2012; Guidolin & Timmermann, 2007). This approach aligns with the economic intuition that financial markets do not move randomly but tend to exhibit persistent phases of relative calm, growth, stress, or crisis that require different investment responses (Ang & Bekaert, 2002).

Different clustering algorithms vary in their suitability for regime detection. For example, k-means is simple but assumes spherical, non-overlapping clusters and equal variance,

assumptions that rarely hold for financial time series (Karim, et al., 2024). Hierarchical clustering can be unstable in noisy data and provides no probabilistic framework for transition dynamics. In contrast, Gaussian Mixture Models (GMMs) have gained traction for this task because they accommodate overlapping clusters and varying covariance structures, aligning well with the continuous and often overlapping nature of financial regimes (Botte & Bao, 2021). GMMs offer soft cluster membership, allowing each time period to have a probabilistic association with multiple regimes. This reflects the reality that regime shifts are rarely abrupt and can evolve gradually, a nuance that hard clustering methods may fail to capture. Density-based methods such as HDBSCAN have also been explored for their ability to detect irregular shapes and isolate outliers. However, benchmark studies demonstrate that HDBSCAN often merges overlapping clusters when data lacks clear density gaps, which is common in smoothly evolving market states (Campello et al., 2015). Taken together, probabilistic methods like GMMs may better capture gradual transitions inherent in financial regimes due to their balance among flexibility, interpretability, and computational tractability, which justifies their use in this study.

2.3. Dimensionality Reduction and PCA

A critical step in making regime identification tractable is dimensionality reduction, and Principal Component Analysis (PCA) remains one of the most established and interpretable tools in this respect (Jolliffe, 2002). PCA not only condenses the dimensionality of a dataset to a smaller number of uncorrelated factors, but also reveals how each original variable contributes to these derived components through its loadings matrix. These loadings, which are specifically the correlation coefficients between original variables and principal components, are often interpreted to identify dominant drivers of variance (Jolliffe, 2002). In empirical finance, loadings can reveal economically meaningful patterns: for instance, a first principal component with uniformly positive loadings across assets may represent a broad market factor, while subsequent components may highlight sector specific or size-based co-movements (Connor & Korajczyk, 1988). Studies argue that examining loadings helps maintain interpretability when selecting features, avoiding the “black box” effect that sometimes plagues the practical value of dimensionality reduction, ensuring that selected features preserve economically relevant signals while discarding noise. Although nonlinear methods like t-SNE and UMAP are helpful for visualizing latent structures, PCA remains a practical standard in finance precisely because its loadings support robust factor interpretation, improving the transparency of regime detection and clustering frameworks (Marx, 2024).

2.4. Modelling Regime Dynamics with Markov Chains

Identifying regimes is only the first step; it is critical to understand their temporal dynamics, because simply labelling states at each point in time does not tell the portfolio manager how likely those states are to persist or change. Markov chains offer a tractable yet powerful framework for modelling both regime persistence, that is the probability of remaining in a given state, and transition probabilities, the likelihood of moving to another regime in the next

time step (Damásio & Nicolau, 2014). This is important because real-world financial markets rarely jump randomly between regimes; instead, they tend to exhibit path dependency and inertia, meaning today's market environment influences tomorrow's.

While simple first-order Markov chains can capture this path dependence in a clear and interpretable way, more sophisticated multivariate Markov-switching models and Markov-switching Vector Autoregressions (MS-VARs) can explicitly account for interactions among multiple variables or asset classes (Damásio & Mendonça, 2018). Recent methodological contributions, such as the *GenMarkov* framework in R, demonstrate how flexible generalized multivariate Markov chains can be practically implemented to model complex regime interactions in an accessible way (Damásio & Vasconcelos, 2022). These richer models extend the basic framework by modelling not just regime persistence but also how interdependencies evolve across regimes. However, they typically require larger datasets and introduce additional estimation and interpretability challenges, which may be impractical when the objective is to keep a regime framework transparent and actionable for portfolio allocation.

The first-order Markov assumption, that the probability of the next state depends only on the current state, is a practical simplification that allows these dynamics to be estimated with relatively modest data requirements and without introducing unnecessary model complexity (Ang & Bekaert, 2002). Once estimated, the transition matrix provides valuable insight into the expected duration of regimes through the self-transition probabilities and typical pathways of regime shifts. This can directly inform forecasting, because knowing that a calm "risk-on" regime has historically been highly persistent suggests a higher likelihood of continuation, whereas a "crisis" regime may have a higher probability of transitioning quickly back to a recovery or neutral state.

From a portfolio management perspective, this temporal structure helps quantify and manage regime change risk, supporting more adaptive tactical asset allocation decisions. For instance, if the probability of shifting into an adverse regime rises, the portfolio can be adjusted proactively to hedge risk or reduce exposure, rather than reacting only after a shift has occurred. While real-world market transitions can be non-stationary and affected by structural breaks (Damásio & Nicolau, 2024), the first-order Markov chain remains a widely accepted approximation, especially when the goal is to estimate regime durability and switching risks in a practical portfolio context.

2.5. Tactical Asset Allocation and Regime Use

Tactical Asset Allocation (TAA) extends the idea that asset return distributions and risk premia, meaning the excess returns earned for taking on additional risk, are time-varying and partially predictable, offering portfolio managers a way to actively tilt exposures in response to changing market conditions. Unlike strategic allocation, which holds fixed weights over the long term, TAA deliberately adjusts portfolio weights over the short to medium term to exploit temporary opportunities or mitigate risks identified through macro signals, valuation spreads,

or market regime insights (Ang & Bekaert, 2002). Empirical evidence suggests that state-dependent or regime-aware strategies can meaningfully enhance risk-adjusted returns and reduce drawdowns if regimes are sufficiently persistent to capture (Faber, 2007). However, the literature also emphasizes that the success of TAA hinges on robust regime detection, credible forecasting, and the ability to implement portfolio adjustments efficiently before adverse transitions gradually destroy performance (Oliveira et al., 2025). Frequent reallocations can introduce significant transaction costs, estimation errors, and timing risks that may offset theoretical gains. Therefore, successful TAA depends on the credibility of the regime model, the persistence of identified states, and the investor's ability to implement changes efficiently, while also balancing flexibility with practical constraints to ensure that the benefits of adapting to changing conditions outweigh the costs. By linking regime probabilities to allocation decisions, this research tests whether a data-driven regime framework can practically enhance the flexibility of an allocation strategy without adding undue complexity.

2.6. BRIDGING THE GAP

Despite advances, gaps remain. Prior studies often isolate regime modelling to a single asset class or rely heavily on macroeconomic indicators, leaving open questions about how different market segments, such as large, mid, and small cap equities and cryptocurrencies, co-evolve across regimes. Moreover, while regime identification and Markov regime-switching have both been well studied, relatively few works fully integrate unsupervised detection with an explicit forecasting step that captures regime persistence and transition risks. Even fewer translate these probabilities directly into a forward-looking allocation framework that systematically adjusts portfolio weights based on regime forecasts.

This thesis addresses these gaps by proposing that: (1) regimes can be objectively detected across a mixed universe of traditional and digital assets via PCA-informed clustering; (2) regime transitions can be quantified using a first-order Markov chain that estimates realistic transition probabilities, transforming static labels into a probabilistic forecast of where the market is likely heading next; and (3) these insights can systematically guide a TAA approach that adjusts portfolio weights in a forward-looking, risk-aware manner. In doing so, this research contributes to the literature by proposing and testing an end-to-end approach that bridges unsupervised learning, Markov regime dynamics, and tactical portfolio management in a mixed-asset context, showing how robust data-driven regime detection can enhance portfolio decisions in a mixed-asset setting, in a way that remains interpretable, tractable and practically actionable.

3. DATA AND METHODOLOGY

In this section, we detail the data and the step-by-step methodology used to carry out the analysis outlined above. The process consists of data collection and cleaning, feature engineering, dimensionality reduction assessment with PCA, clustering with GMM, construction of the Markov transition matrix from the clustering results, and development of the regime-weighted tactical asset allocation strategy. We also highlight practical decisions made along the way, such as handling non-trading days and selecting rolling windows, and the reasoning behind those choices.

3.1. ASSET UNIVERSE

The dataset used in this study consists of daily financial data for multiple asset classes over a period of several years, with data gathered from Investing.com. Specifically, the data included a mixed set of 10 assets spanning equities and cryptocurrencies to capture a broad spectrum of market dynamics. These include large-cap U.S. equities (SP500 - the S&P 500 index, also proxied by VV - Vanguard Large-Cap ETF), mid-cap U.S. equities (IJJ - iShares S&P 400 Mid-Cap Value ETF, and IMCV - iShares Morningstar Mid-Cap Value ETF), small-cap U.S. equities (IWM - iShares Russell 2000 ETF), international developed-market equities (EFV - iShares MSCI EAFE Value ETF), emerging market equities (EEM - iShares MSCI Emerging Markets ETF), a sector-focused U.S. equity fund (IYJ - iShares U.S. Industrials ETF), and two major cryptocurrencies (BTC - Bitcoin, and ETH - Ethereum). This diverse set constitutes mixed asset classes, combining traditional stocks with digital assets to enrich the regime analysis. The period of study is March 2016 through December 2024, encompassing varied market conditions, that is the post-2016 equity bull run, volatility shocks in 2018, the 2020 pandemic crash, the 2021 crypto boom, and the 2022-2023 inflationary environment.

To illustrate the basic characteristics of the assets in the sample and to provide context for their risk-return profiles, the table below (Table 1) provides descriptive statistics of daily returns for each asset in the sample. This includes mean return, standard deviation (volatility), minimum and maximum returns, skewness, and kurtosis. This summary highlights the diversity of return distributions and volatilities across traditional equities and cryptocurrencies, which is central to the regime analysis that follows.

Table 1: Descriptive statistics for daily returns of the selected asset universe (2016–2024)

Assets	Mean Daily Return	StD of returns (Volatility)	Minimum Return	Maximum Return	Skewness	Kurtosis	Numbers of observations
SP500	5.50	112.47	-1094.24	906.03	-0.60	13.60	2218
IMCV	0.55	189.52	-6624.71	1199.61	-19.41	674.51	2218
IJJ	-31.61	446.57	-5669.00	1105.00	-10.06	110.11	2218
IWM	4.34	145.34	-1326.69	914.91	-0.60	8.14	2218
EEM	2.02	129.25	-1247.92	805.29	-0.63	9.03	2218
VV	5.55	114.23	-1198.88	972.61	-0.59	15.21	2218
IYJ	5.11	125.26	-1246.23	1230.80	-0.39	14.31	2218
EFV	1.47	116.00	-1176.62	987.42	-1.21	16.58	2218
BTC	26.33	399.34	-3918.00	2556.00	-0.14	7.55	2218
ETH	28.96	536.41	-4455.00	2951.00	0.14	5.42	2218

3.2. Alignment and Cleaning

To ensure a clean, consistent dataset, we deliberately selected U.S.-based equity indices that operate on a broadly shared holiday calendar, minimizing the risk of frequent mismatches that would require removing excessive data during alignment.

However, an additional challenge arises when combining traditional equity indices, which operate only on weekdays, with cryptocurrencies, which trade continuously including weekends. To address this, we decided to explicitly exclude weekend observations entirely and restricte the data to weekdays-only, rather than to forward-fill index prices through weekends to match crypto data, as doing so would introduce unnecessary noise and artificial patterns into the dataset. This choice ensures that all assets are aligned on the same trading-day calendar and prevents distortions in return calculations. By preserving only Monday-Friday trading days for all assets, we established a consistent timeline across datasets and prevented distortions in return calculations. Moreover, since the analysis later applies rolling windows and moving averages, the exact daily crypto prices on weekends are not critical, making this exclusion methodologically appropriate. For occasional non-trading days due to regional holidays, if an asset did not trade while others did, we removed that day across all

series to maintain continuity. All anomalies and outliers were careful checked, and genuine market extremes were retained because they are crucial for identifying regimes.

After aligning dates, we carried out a comprehensive preprocessing pipeline to clean and transform the raw data. First, all column labels were standardized and any extraneous symbols were removed. For example, price columns originally containing currency symbols or percentage signs were renamed to simple identifiers (e.g. “Close”, “Volume”) and stripped of non-numeric characters. Volume figures, which were sometimes abbreviated with suffixes (such as “K” for thousands or “M” for millions), were converted into pure numeric values representing the actual traded volume. Next, any missing data points were imputed carrying the last available observation forward. This forward-filling technique was particularly important after removing weekends and holidays, to ensure continuity in the time series and to avoid introducing blanks in rolling window calculations. We verified the integrity of the price data by recalculating daily returns from the adjusted close prices and confirming consistency with any reported return series. This validation step ensured that no anomalies (such as sudden jumps due to data errors or corporate actions) were present, or if they were, that they were understood and handled appropriately. At the end of this preprocessing stage, we had a clean, aligned dataset of daily observations for each asset, with uniform structure and ready for feature engineering. All assets, whether equity or crypto, could now be analyzed on equal footing over the same period, 2016-2024, without calendar inconsistencies or formatting issues.

3.3. FEATURE ENGINEERING

With the data properly aligned and cleaned, we engineered a variety of features to characterize the behavior of each asset over time. We focused on rolling-window statistics that capture recent performance, risk, and distributional shape of returns. Using a fixed 10-days rolling window, we computed for each day and each asset the rolling volatility, Sharpe ratio, moving average returns, and higher moments such as skewness and kurtosis of returns.

Volatility was defined as the rolling standard deviation of daily returns within the window, serving as an estimate of short-term risk or uncertainty. The daily Sharpe ratio was calculated using the window’s average daily return divided by its standard deviation, and was then annualized, scaling the daily Sharpe by $\sqrt{252}$ to reflect reward-to-risk on an annual basis. The rolling mean return, a simple moving average of daily returns over the window, provides a glimpse of momentum. The rolling skewness and kurtosis were computed to measure the asymmetry and tail heaviness of the return distribution in the window, respectively. Skewness indicates whether the distribution of returns is biased to the upside or downside, with positive skew implying a long right tail of gains, and negative skew indicating frequent small gains and rare large losses, while kurtosis reflects the fat-tailed nature of returns, with high kurtosis corresponding to more extreme outliers than a normal distribution would predict.

All the afformentioned features together form a comprehensive description of each asset's recent behavior: volatility and Sharpe capture risk-adjusted performance, the mean return captures trend, and skewness/kurtosis capture deviations from normality.

3.4. Dimensionality Reduction and Feature Selection using PCA Loadings

Before proceeding to cluster the data, we applied dimensionality reduction to understand the structure of the feature set and to guide feature selection. Working with one of the ten datasets as the representative one, all the engineered features were first standardized to ensure that they are on comparable scales, since features like volatility or volume could have numeric ranges vastly different from, say, skewness. We then performed Principal Component Analysis (PCA) on the combined standardized feature set. PCA allowed us to identify linear combinations of the features (principal components) that explain the most variance in the data. By examining the variance explained by each component, we determined an appropriate number of dimensions to retain for effective data representation.

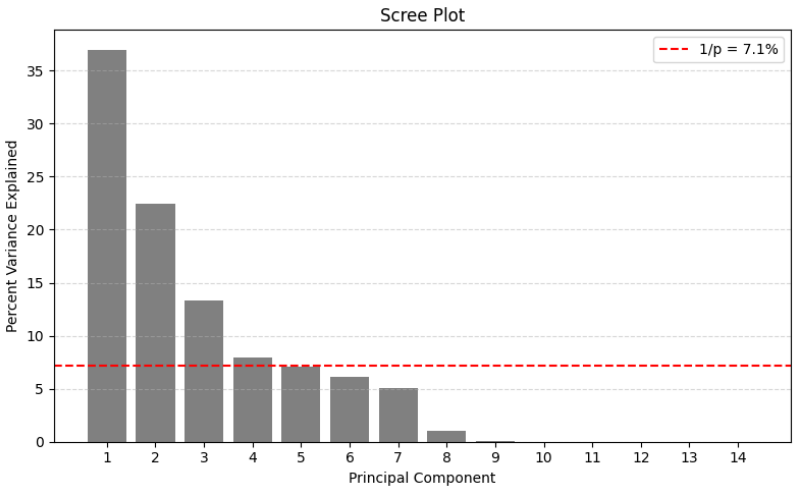

Figure 1 - Scree plot of PCA showing the percentage of variance explained by each principal component

The scree plot displays a declining curve, where the first few principal components account for a large fraction of the total variance, and each subsequent component contributes progressively less. In this plot, we see a pronounced elbow in the curve: the first principal component explains a significant portion of variance indicating that many of our features are correlated or capture related information. Components two and three also provide substantial information, but beyond a certain index, the additional variance explained by each new component diminishes sharply. The horizontal reference line marked around 7% corresponds to the equal share of variance if all components were equally important at 14 components, i.e. 1÷14 is approx 7.1%. Components above this line contribute more than the average, whereas those below contribute less. In our case, the scree plot indicated that the first three or four principal components would capture most of the informative variance in the data, while later components were mostly capturing noise or very minor patterns.

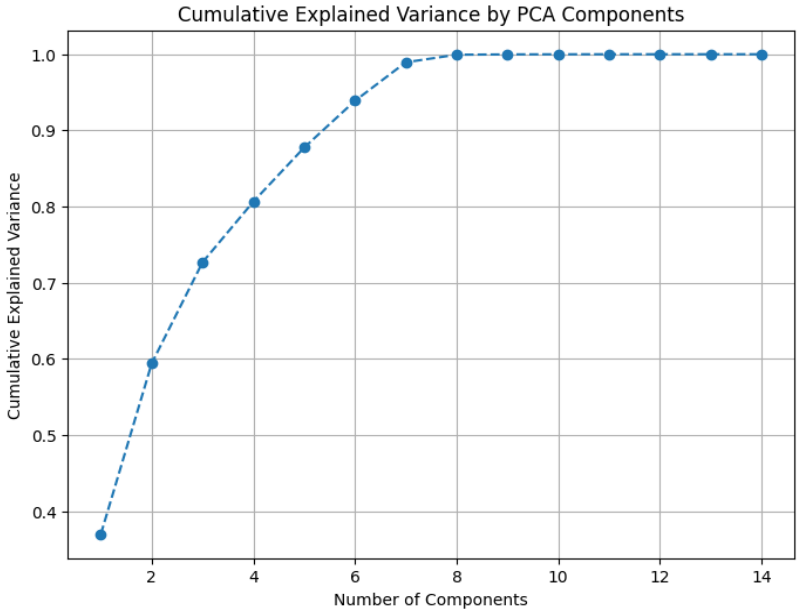

Figure 2 - Cumulative explained variance as a function of the number of PCA components

The above plot illustrates how the total variance explained grows as we include more principal components. It rises quickly with the first few components and then asymptotically approaches a steady level. We observe that a large majority of the variance in the dataset is captured by the time we include around five to six components, and by the time we reach about eight components the cumulative explained variance is essentially saturating near 100%. This means that the remaining components beyond the eighth contribute very little additional explanatory power. Based on this analysis, it has been decided that retaining the top four oto six principal components would be sufficient for our purposes. The exact components to retain will be finalized by further analyzing the loadings table.

Instead of performing clustering directly in the reduced dimensional principal component space, we examined the principal component loadings, which describe the correlation structure between the original variables and the principal components. This analysis highlighted which features contributed most significantly to the variance in the data and thus carried the most informative value for subsequent analyses.

Specifically, the first principal component was found to represent a general “price trend” factor, with high loadings from variables of Price, Open, High, Low, and the Simple Moving Average (SMA). Accordingly, only ‘SMA’ was retained as a representative due to the high collinearity among these measures. The second component captured aspects of return dynamics and risk-adjusted performance, leading to the selection of ‘Annualized_Sharpe_ratio’ and ‘Mean_Changepct’ to reflect return quality and smoothed price change, respectively. The third component predominantly represented risk, with ‘Annualized_Volatility’ chosen for its strong loading. The fourth component was associated with the distributional characteristics of returns, warranting the inclusion of ‘Skewness’ and ‘Kurtosis’. Additionally, the fifth component captured a volume related dimension, justifying the retention of the ‘Volume’ metric, while the sixth component, which reflected short-term

return noise, did not necessitate additional features since 'Mean_Changepct' was already selected. The above observations were captured from PCA loadings table as presented in Table 2.

Table 2: PCA Loadings Table - feature contributions to each component

	PC1	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**
Price	0.4327	0.0688	0.0813	0.0362	-0.0153	-0.0260
Open	0.4335	0.0582	0.0839	0.0426	-0.0158	0.0125
High	0.4328	0.0639	0.0887	0.0385	-0.0154	-0.0058
Low	0.4335	0.0645	0.0737	0.0402	-0.0147	-0.0111
Volume	0.0283	0.0208	0.0417	-0.0734	0.9945	0.0100
Changepct	-0.0363	0.2391	-0.0173	-0.1547	-0.0077	-0.9553
Volatility	-0.1447	0.0975	0.6790	0.0036	-0.0226	0.0143
Annualized_volatility	-0.1447	0.0975	0.6790	0.0036	-0.0226	0.0143
Mean_Changepct	-0.0686	0.5225	-0.0270	-0.0645	-0.0183	0.1243
Sharpe_ratio	-0.0389	0.5387	-0.1021	-0.0675	-0.0182	0.1676
Annualized_Sharpe_ratio	-0.0389	0.5387	-0.1021	-0.0675	-0.0182	0.1676
SMA	0.4335	0.0331	0.0949	0.0378	-0.0106	-0.0177
Skewness	-0.0469	0.2291	-0.1272	0.5732	0.0827	-0.0748
Kurtosis	-0.0887	0.0020	0.0462	0.7881	0.0302	-0.0913

Based on these insights, we identified a subset of features that collectively span the key dimensions of variation in the data, ensuring that redundant or uninformative variables were excluded; for example, highly collinear features or those contributing primarily to low-variance components were omitted to avoid unnecessary complexity. This feature selection process determined the final set of features: ['SMA', 'Annualized_Volatility', 'Annualized_Sharpe_ratio', 'Mean_Changepct', 'Volume', 'Kurtosis', 'Skewness']. Once this reduced feature set was determined, all asset datasets were merged into a single combined dataset containing the selected features for every asset-day observation. The data were then standardized after merging to maintain consistent scaling, since merging can slightly shift overall means and variances. At this stage, we have a unified matrix of observations ready for

clustering. The PCA analysis thus served a dual purpose: it helped reduce dimensionality and noise, and it provided insight into which features are most important, thereby informing our feature selection prior to clustering.

3.5. Gaussian Mixture Model clustering

3.5.1. Data Structure

Financial time series data are inherently continuous in nature. While this property is well established in the literature, we further validated the underlying structure by applying the Uniform Manifold Approximation and Projection (UMAP) technique as an exploratory step. UMAP, a nonlinear dimensionality reduction method, was used to project the high-dimensional feature space into two dimensions to visualize the distribution of data points. The resulting projection confirmed that the data points exhibit significant overlap, with no clear boundaries that would indicate distinctly separated clusters.

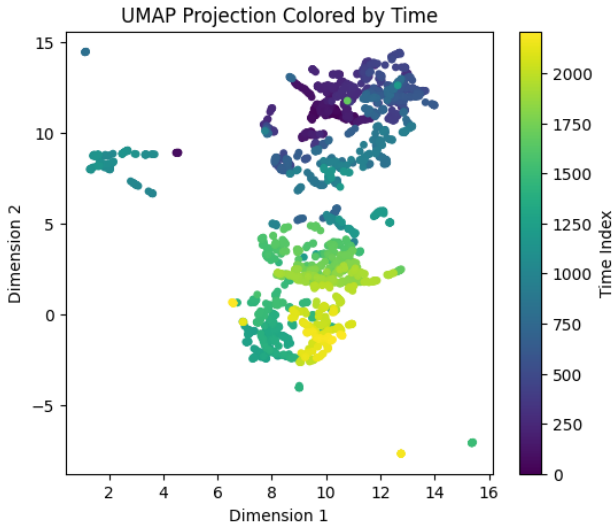

Figure 3 - UMAP projection colored by time

This observation further reinforces the appropriateness of employing a Gaussian Mixture Model (GMM) for regime identification, as GMMs are well-suited to modeling overlapping, continuous distributions and assigning probabilistic cluster membership. By design, a GMM assumes that the data arises from a mixture of Gaussian components that can overlap, which accommodates the smooth transitions often present in financial time series. In contrast, hard clustering methods such as k-means can produce arbitrary splits in a continuous space, forcing points into discrete clusters even when boundaries are unclear. Similarly, density-based approaches like HDBSCAN may struggle to separate regions without clear, low density gaps, potentially merging the majority of the data into a single cluster or misclassifying transitional points as noise. The soft assignment mechanism of GMM, which allows each observation to belong to multiple clusters with varying probabilities, aligns well with the gradual regime shifts observed in the market data. Therefore, the visual evidence of overlapping structure strongly

supports the choice of a probabilistic mixture model to capture the nuanced, continuous nature of market regimes in this study.

3.5.2. Clustering

Having decided on using Gaussian Mixture Models for clustering, the next step was to determine how many clusters (components) would be appropriate for the data. Choosing the number of clusters is a critical model selection problem, as too few clusters might oversimplify by lumping distinct market conditions together, while too many clusters could overfit noise and create spurious distinctions. We approached this problem by examining statistical fit metrics across a range of possible cluster counts, and by considering the interpretability of the results.

While silhouette scores are commonly used to assess clustering quality, they assume that clusters are well-separated and ideally convex in the feature space. However, in time-evolving financial regimes, transitions are often gradual and overlapping. As shown in the UMAP projection colored by time (Figure 3), the data does not form compact, isolated clusters but rather follows a smooth temporal evolution. Therefore, silhouette scores may undervalue clustering solutions that are otherwise meaningful in economic terms. In this context, model-based selection criteria such as the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC) are more appropriate, as both assess the trade-off between model fit and complexity within the probabilistic framework of the Gaussian Mixture Model. BIC applies a stronger penalty for additional parameters, favoring simpler models when evidence for extra clusters is weak, while AIC tends to reward improved fit even if the complexity slightly increases (Fraley & Raftery, 2002). Finally, interpretability and alignment with plausible market regimes provide additional practical justification for selecting the number of clusters that best captures distinct market conditions.

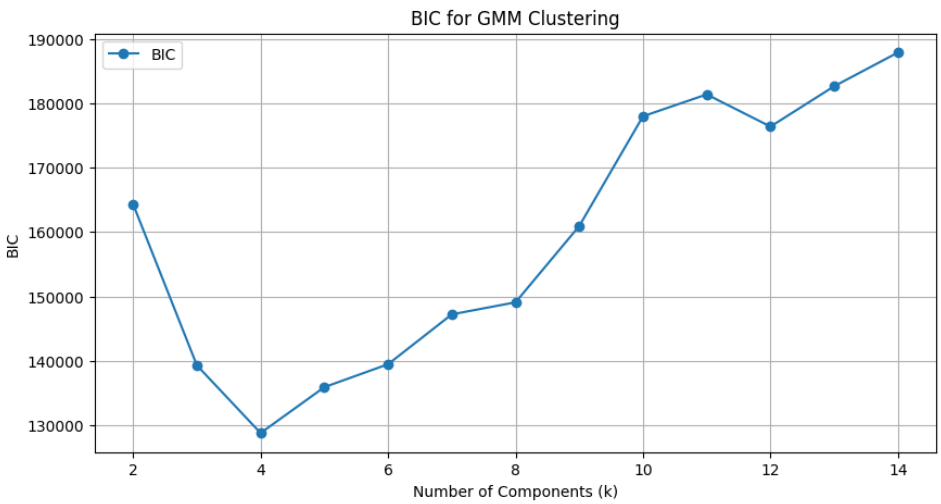

Figure 4 - BIC values for GMM models with varying numbers of clusters

The BIC is a penalized likelihood criterion, with lower BIC indicates a better fit when balancing model complexity against goodness of fit. In the plot above, we can see how the BIC changes as we increase the number of Gaussian mixture components. Initially, adding more clusters dramatically improves the fit (BIC decreases sharply), but beyond a certain point of four components, the improvements level off and the BIC curve starts to flatten and even turn upwards. In our analysis, the BIC reached its minimum at four clusters. This suggests that a GMM with four components provides the best trade-off: it fits the data significantly better than two or three component models, but adding a fifth cluster yields only negligible improvement and in fact BIC begins to rise indicating extra complexity.

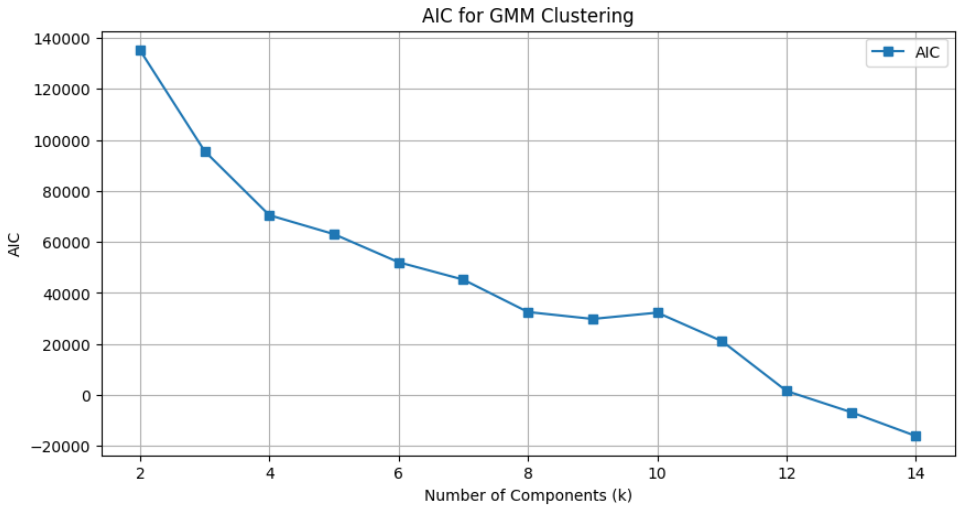

Figure 5 - AIC values for GMM models with varying numbers of clusters

We also looked at the Akaike Information Criterion (AIC), which is similar to BIC but penalizes complexity less strictly. The AIC curve continued to decrease as more clusters were added, however, the rate of decrease in AIC slowed beyond three or four clusters, mirroring the point of diminishing returns indicated by BIC. Together, these information criteria suggested that the model's fit improvement beyond four clusters was not substantial, while three clusters also seemed capable of capturing the major structure in the data without unnecessary complexity.

To ensure robustness, both the three and four cluster solutions were further evaluated in terms of economic interpretability, statistical characteristics, and practical relevance for regime identification. In conventional regime-switching frameworks, financial markets are typically segmented into three broad states: bull (risk-on), neutral (calm), and bear (risk-off). This trichotomy has long been favoured in both academic literature and industry practice for its intuitive simplicity and ease of communication. However, this standard approach can fail to distinguish between qualitatively different risk-off environments, such as mild corrections versus severe crises, or between neutral phases that may subtly differ in market participation, liquidity, or cross-asset dynamics.

By adopting a four-cluster Gaussian Mixture Model instead, this analysis explicitly captures such nuances, moving beyond the classic bull/neutral/bear structure. The four regimes can be summarized as follows:

1. **Cluster 1 - Risk-On/Strong Bullish Regime**: This cluster captures a Strong Bull Market Regime, where risk-adjusted returns are consistently high, momentum is positive, and volatility is moderately elevated, typical of post-crisis rallies. Return distributions often show mild positive skew, indicating upside opportunities. Trading volumes are neutral to slightly elevated, supporting broad participation.
2. **Cluster 2 - Crisis/Severe Risk-Off Regimes**: This regime captures full-blown crisis periods marked by high volatility, strongly negative returns, and significantly negative Sharpe ratios across all major assets. Trading volumes are generally above average, consistent with panic-driven selling and forced de-risking. This regime reflects periods of broad-based panic or systemic shocks, comparable to market conditions during the 2008 financial crisis or the initial COVID-19 crash.
3. **Cluster 3 - Calm/Neutral Regime**: This regime reflects a quiet or sideways market phase with low volatility and modest positive returns. Risk-adjusted performance is slightly positive, and kurtosis and skewness are near zero, implying a stable environment with normal return distributions. Investor activity tends to be neutral or slightly below average, consistent with consolidation periods or early recoveries where optimism is still building.
4. **Cluster 4 - Moderate Turbulence/Soft Risk-Off Regime**: This regime lies between calm markets and severe crisis, with strongly positive trends but muted momentum, suggesting a disconnection between elevated price levels and underlying investor persuation. Volatility and drawdowns are elevated, capturing a fragile state environment. Below average volumes often signal cautious repositioning rather than outright panic.

Crucially, the distinction between the Crisis and Moderate Turbulence regimes (Clusters 2 and 4) demonstrates the benefit of moving beyond three clusters. While both represent risk-off conditions, their severity and market characteristics differ meaningfully. Cluster 2 reflects full-scale panic with sharp drawdowns and liquidity stress, whereas Cluster 4 indicates a deterioration phase or soft correction that may precede a crisis but has not yet escalated into outright market collapse. Capturing these intermediate states provides early signals of market stress and allows for more responsive, nuanced portfolio management.

In conclusion, although both the three and four cluster solutions are statistically defensible, the four-regime framework better reflects the empirical reality of complex, transitional market behaviour. This approach supports more detailed monitoring of regime shifts and avoids the limitations of the traditional bull/neutral/bear model, ultimately enhancing the capacity for informed risk management and dynamic allocation strategies.

3.6. MARKOV CHAIN - BASED REGIME TRANSITION MATRIX CONSTRUCTION

With the asset clusters (market regimes) identified by the Gaussian Mixture Model, we next model the dynamics of regime shifts using a first-order Markov chain. A Markov chain is a stochastic process that transitions between states with certain probabilities. The key assumption is the Markov property, meaning the probability of moving to the next state depends only on the current state and not on the sequence of prior states.

Formally, let s_t denote the sequence of latent regimes, where each regime at time t is defined as $s_t \in \{1, 2, \dots, K\}$. Note that in this study, K is equal to four. The Markov property holds:

$$P(s_{t+1} = j \mid s_t = i, s_{t-1}, \dots, s_1) = P(s_{t+1} = j \mid s_t = i) \ \forall \ i, j.$$

This implies that given the present market regime, the likelihood of entering any particular next regime depends only on the present regime, regardless of the historical path taken to the current regime. This approach is suitable for financial time series that exhibit recurring regimes, as it allows the characteristics of the market to change when shifting from one regime to another.

To implement the Markov regime-switching model, we constructed a transition probability matrix P based on the historical sequence of cluster labels obtained from the GMM clustering. Each unique cluster label is treated as a state in the Markov chain, so the full time series of assigned clusters forms the observed sequence of regimes.

Formally, the transition matrix is defined as:

$$P = [P_{ij}] \, , \ P_{ij} = P(s_{t+1} = j \mid s_t = i) \, .$$

The transition probabilities are estimated empirically by counting how often the system moves from regime i to regime j from one time step to the next. In practice, this is implemented by iterating through the entire sequence of regime labels and recording each transition from state i to state j. For each observed transition, we increment a count in a $K \times K$ matrix, whose rows and columns correspond to the four regimes. This counting step results in a raw frequency matrix that shows how many times each type of transition occurred in the sample. For this, we define the empirical count matrix:

$$\mathrm{N_{ij}} = \sum_{t=0}^{T-1} 1 \, (\, s_t = i \, , \, s_{t+1} = j \,),$$

where for each pair of consecutive observations (s_t, s_{t+1}), the indicator function $1(\cdot)$ equals one when the condition is true.

Next, the matrix is normalized by dividing the counts in each row by the total number of transitions originating from that regime, so that each row sums to one. This ensures that each row of the transition matrix represents a valid probability distribution conditional on the

current regime. Precisely, the estimated transition probability for moving from state i to state j is given by:

$$\widehat{P_{ij}} = \frac{N_{ij}}{\sum_{j=1}^{K} N_{ij}} \ , \qquad with \sum_{j} \widehat{P_{ij}} = 1 \, .$$

This step transforms raw transition counts into conditional probabilities, which aligns with the Markov property: given the current regime, the row shows the likelihood of staying in the same regime or moving to another one. The resulting transition probability matrix offers clear insights into regime dynamics, and will be further analysed in the next chapter.

To assess the reliability of the estimated Markov transition matrix and its predictive validity, an out-of-sample log-likelihood and perplexity analysis was conducted. Specifically, the full sequence of regime labels was split into a training set used to estimate the transition probabilities, and a test set to evaluate model fit, of 80-20% split respectively. For each pair of consecutive states in the test set, the log-likelihood was calculated based on the corresponding probability from the trained transition matrix, with a small smoothing constant *epsilon* added to prevent zero-probability issues. The average log-likelihood per transition was then used to compute the perplexity:

$$Perplexity = \ e^{-Average\ Log-Likelihood}.$$

Perplexity quantifies the model's uncertainty: lower perplexity indicates stronger predictive power and greater certainty about regime persistence and switching. In this study, the resulting perplexity for the test set was approximately 1.11. Given that four regimes were modeled, this low perplexity value reinforces the finding of strong regime persistence, making the out-of-sample surprises few. However, if the market were to experience structural breaks, for example, extreme events that occur only once during the sample, the test set could contain regime transitions that were not present in the training set, lowering the log-likelihood and raising perplexity. This is not a model flaw, but rather an honest reflection of the challenges in forecasting regime shifts during rare or unprecedented market conditions

This pattern confirms that the market conditions captured by the clustering are stable enough over time to justify using a first-order Markov chain for forecasting. By quantifying how likely it is for the market to remain in the current regime or transition to another, the transition matrix translates the static regime classification into a dynamic, forward-looking forecast. This provides a robust input for the tactical asset allocation strategy developed in the next step and directly supports more adaptive portfolio decisions.

3.7. REGIME-SWITCHING TACTICAL ASSET ALLOCATION

The final step of this methodological framework culminates with the tactical asset allocation process, which brings together the preceding analysis to construct a forward-looking portfolio that adapts dynamically to changing market conditions. Asset allocation serves as the practical

implementation of the regime-switching framework, integrating the identification of distinct market states and their respective risk-return characteristics into actionable portfolio weights. It thus represents the point at which the entire analytical pipeline, from regime identification to transition probability estimation, is translated into a concrete investment decision. This step ensures that the insights gained from the detection of regimes and estimation of their transition dynamics directly inform how capital is allocated among assets, completing the link between the research question and its practical application.

In the broader context of portfolio theory, the objective of an investor naturally shapes the focus of asset allocation. For example, an investor whose primary goal is to minimize total portfolio risk may employ a minimum variance approach. Alternatively, if portfolio stability is important, practical measures of penalizing large deviations from existing positions and applying robust constraints on position limits may be incorporated to smooth portfolio adjustments and maintain more consistent allocations over time. For this study, the Sharpe ratio is selected as the central optimization objective. The rationale behind this choice lies in the Sharpe ratio's balanced nature, as it explicitly considers both expected return and risk. This approach helps avoid the pitfalls of pursuing higher returns at the cost of excessive volatility, or, conversely, minimizing risk so cautiously that valuable opportunities for return are lost. By targeting the highest risk-adjusted return, the approach aligns with a pragmatic and widely adopted standard in asset management, ensuring that the resulting portfolios reflect an optimal tradeoff between reward and uncertainty within each regime.

3.7.1. Implementation

Having identified distinct market regimes and estimated their transition dynamics, this study implements these insights through a structured tactical asset allocation process that systematically adapts the portfolio to changing market conditions. The core idea is that each regime reflects a different set of market characteristics, which may justify holding a different "optimal" mix of assets. By estimating the probability of moving between regimes, it becomes possible to combine these regime-specific portfolios into a single, forward-looking allocation that is robust to market uncertainty.This process can be understood in three practical steps.

First, for each identified regime, the historical data is divided to create regime-specific samples of daily asset returns. From these, the model estimates the expected return vector and the covariance matrix of returns for each asset. These estimates form the inputs for the portfolio optimization step.

Second, for each regime, an optimization problem is solved to find the portfolio weights that maximize the Sharpe ratio, that is the expected excess return per unit of risk. This is done under realistic constraints: the portfolio weights must sum to one, remain non-negative, and are capped at a maximum weight per asset to avoid excessive concentration. The optimization is performed numerically using a Sequential Least Squares Programming (SLSQP) method,

which handles the nonlinear objective and constraints reliably. This step generates a distinct regime-specific optimal portfolio, reflecting its unique risk-return profile.

Third, the model combines these regime-specific portfolios using the estimated regime transition matrix. As previously noted, this matrix quantifies the probability of moving from the current regime to each possible regime in the next period. The final tactical allocation is then calculated as a probability weighted average of all regime portfolios. This approach ensures that the portfolio is not overly dependent on any single regime prediction, but instead reflects the full range of possible market states, making the allocation more adaptive and forward-looking. In practice, this means that each day, the current regime is detected based on the most recent market data, and the transition probabilities are applied to blend the regime-specific optimal portfolios accordingly. This produces a precise vector of expected asset weights for the next trading day, ensuring that the portfolio dynamically adjusts in real-time to reflect the probabilistic nature of regime shifts.

Taken together, this methodological framework ensures that regime identification, transition probability estimation, and portfolio construction are not treated in isolation but instead operate as an interconnected system. Additionally, it translates the earlier analysis of market regimes into concrete asset allocation decisions that directly address the research question of how to adjust portfolios dynamically as market conditions evolve. This foundation provides a robust basis for the subsequent empirical study and discussion of results, which will evaluate the performance and practical implications of the regime-switching tactical asset allocation strategy in real-world conditions.

4. RESULTS AND DISCUSSION

This chapter presents and interprets the main findings derived from the regime-switching tactical asset allocation framework developed in this study. The results are discussed in the context of the initial objective of identifying and forecasting financial market regimes in order to inform portfolio decision-making.

The results of the regime detection stage demonstrate that the financial market can indeed be partitioned into distinct, interpretable regimes, each exhibiting clearly differentiated statistical properties. For example, the cluster-specific descriptive statistics show that Regime 1 is characterized by more moderate volatility with relatively higher Sharpe ratios for equities such as the SP500, mid-cap ETFs, and selected international exposures. In contrast, Regime 2 reflects adverse conditions with negative average returns and heightened annualized volatility across multiple asset classes, resulting in predominantly negative Sharpe ratios. Regime 3 represents a more neutral or transitional market state, marked by modest positive average returns, relatively low volatility, and moderate risk-adjusted performance. Finally, Regime 4 appears to capture expansionary conditions with strongly positive simple moving averages, positive mean returns, and generally lower downside risk, particularly for large-cap equities and industrials. The accompanying boxplot of the SP500's annualized Sharpe ratios across these regimes clearly illustrates these differences in risk-adjusted performance: Regime 1 stands out with consistently higher Sharpe ratios, Regime 2 shows a pronounced concentration of negative Sharpe ratios, and Regimes 3 and 4 cluster around more neutral or mixed values. Taken together, this visual evidence reinforces the economic intuition that market conditions evolve nonlinearly and that certain risk factors tend to cluster together, underscoring the relevance of linking regime detection to tactical portfolio adjustments.

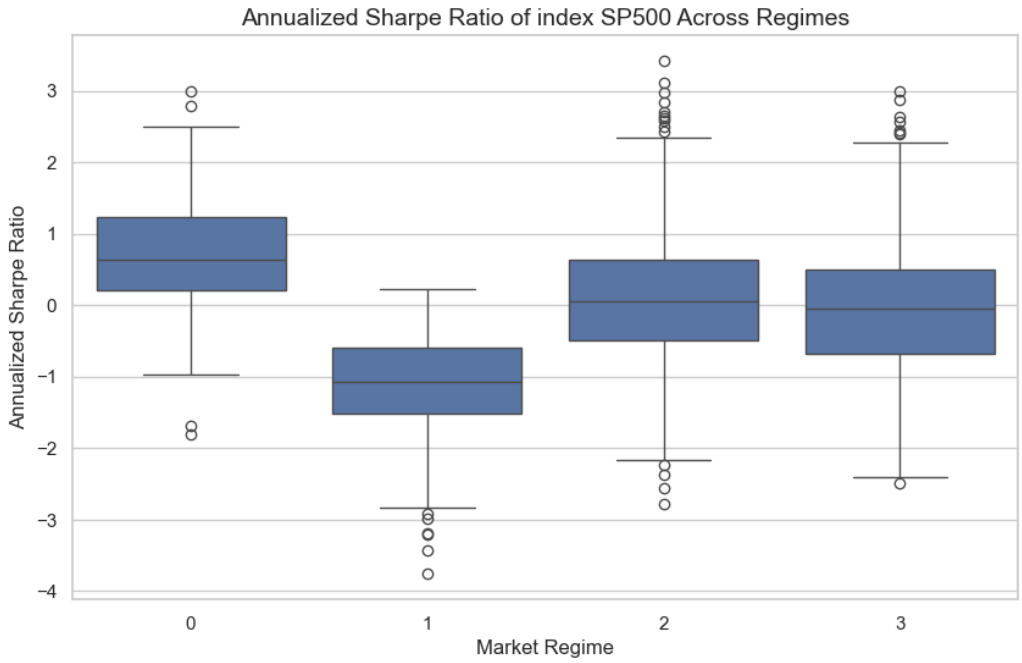

Figure 6 - Distribution of annualized Sharpe ratios for the S&P 500 index across the four identified market regimes

To illustrate how the identified regimes evolve over time, Figure 7 plots the full time series of regime assignments, each period colour-coded by regime. This visual clearly shows phases of stability and points of transition, demonstrating that the regimes are not only statistically distinct but also economically meaningful and persistent for extended periods. For instance, the plot confirms that during the early 2020 COVID-19 crisis, the model correctly identifies multiple sharp spikes into Regime 2, the severe risk-off state. This aligns with extreme market panic, high volatility, and broad-based drawdowns consistent with forced de-risking, exactly what the crisis cluster represents. In contrast, the prolonged stretch of Regime 4 from end-2020 through 2021 illustrates how the framework captures the supportive rebound phase that followed the initial pandemic crash. Earlier in the sample, the plot shows significant periods in Regime 3, the calm or neutral regime, during relatively stable market phases like 2017 and parts of 2019 when volatility was low, and risk-adjusted returns hovered around modestly positive levels. Finally, the more scattered presence of Regime 1, the risk-on or strong bull market regime, appears in concentrated bursts, especially during short-lived rallies or recovery spurts. For example, pockets of Regime 1 are visible following corrections in late 2018 or around mid-2023, consistent with periods where market momentum and risk appetite temporarily surged above normal levels. Overall, these patterns confirm that the detected regimes reflect real economic shifts and market structures rather than arbitrary statistical groupings. They also demonstrate the framework's ability to capture both abrupt transitions, such as crisis shocks, and more gradual shifts as markets move between phases of risk-on sentiment and caution.

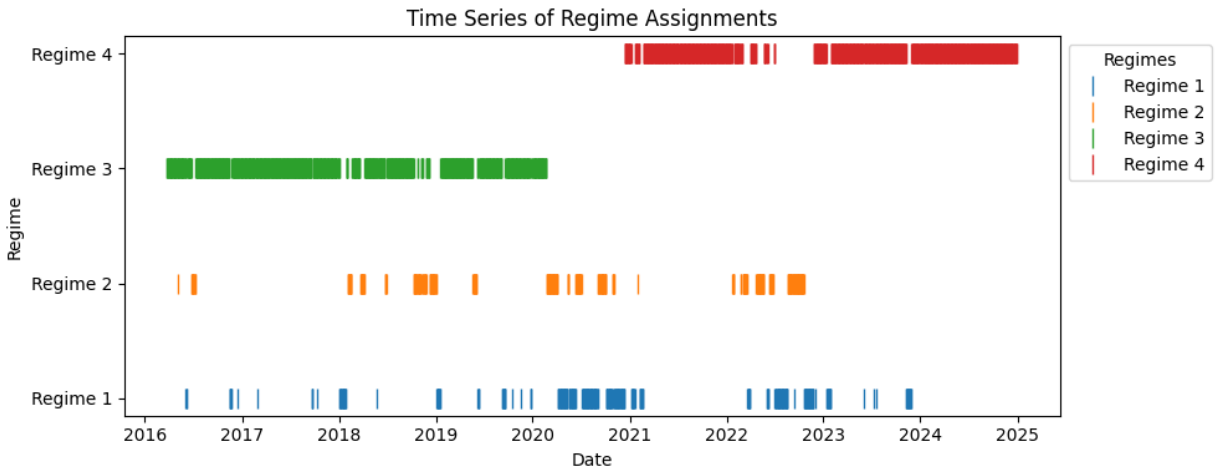

Figure 7 - Time series plot of regime assignments across the study period

Building on this regime segmentation, the study integrates a Markov transition matrix to capture the probabilistic nature of regime shifts. This addition ensures that the asset allocation is not static but adjusts dynamically to form a view of how the market may evolve in the near future. Such dynamic modelling respects the reality that market conditions do not

change abruptly but rather evolve gradually, which is reflected in the probability-weighted approach to portfolio construction. In this study, the estimated matrix shows that each regime is highly persistent. Specifically, Regime 1 has an 89.6% chance of staying in the same state in the next period, Regime 2 has 91.1%, while Regime 3 and Regime 4 have 97.1% and 98.0% persistence respectively. Only a small probability mass is allocated to transitions to other regimes, indicating that regime shifts do occur but tend to be infrequent and gradual rather than abrupt.

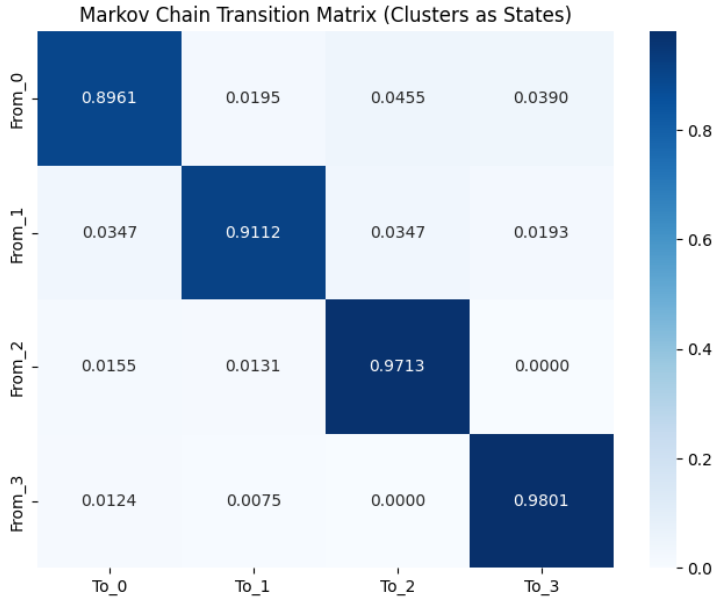

Figure 8 - Heatmap of the estimated regime transition probabilities

This aligns with the observed stability in the time-series plot and provides an empirical basis for forecasting regime durations. In practice, this ensures that the tactical asset allocation is not only regime-aware but also forward-looking.

To illustrate the practical impact of this regime-switching framework, the study computes regime-specific optimal portfolios that maximize the Sharpe ratio while respecting realistic constraints. In this context, the optimized portfolio weights for each regime provide insight into how investors might adjust exposure across different asset classes, including large-cap equities, mid-cap value ETFs, industrials, international equity factors, and cryptocurrencies. The results indicate that during periods corresponding to Regime 1, which has been historically marked by moderate volatility and positive average returns, the optimal tactical allocation places significant weight on diversified small-cap and emerging market equities such as IWM (Russell 2000) and EEM (MSCI Emerging Markets), aligning with the market's relatively supportive conditions for risk assets. In contrast, the optimal allocation for Regime 2, which historically shows negative excess returns and high volatility, emphasizes more defensive positioning, strongly favouring IWM and EFV (developed markets ex-US value stocks), which have historically demonstrated resilience or relative outperformance in turbulent conditions, while avoiding exposure to assets with historically negative Sharpe ratios

in that state. For Regime 3, which represents a more neutral or transitional state with moderate risk-adjusted returns and relatively low volatility, the optimal portfolio shifts significant weights toward mid-cap equities like IMCV and IJJ, while still maintaining limited allocations to large-cap exposures and crypto assets to preserve flexibility. Finally, Regime 4, which reflects expansionary market conditions with strongly positive trends and generally lower downside risk, shows optimal allocations that favour large-cap growth assets like IMCV and IJJ, complemented by moderate positions in emerging markets and selective crypto exposures such as BTC and ETH, which benefit from the supportive environment for higher beta assets. Across most regimes, the optimized weights for Bitcoin and Ethereum are near zero, suggesting that within the studied period, their addition did not consistently enhance the portfolio's risk-adjusted performance. This empirical result is notable given the ongoing debate in the literature regarding the diversification benefits of cryptocurrencies.

Table 3 - Regime-specific optimal portfolios

	Regime 1	**Regime 2**	**Regime 3**	**Regime 4**
SP500	0.0000	0.0000	0.0000	0.1074
IMCV	0.0000	0.0000	0.4000	0.4000
IJJ	0.0000	0.0000	0.4000	0.4000
IWM	0.4000	0.2000	0.0000	0.0000
EEM	0.2626	0.4000	0.0000	0.0000
VV	0.2229	0.0000	0.0684	0.0000
IYJ	0.0933	0.0000	0.0000	0.0000
EFV	0.0000	0.4000	0.0000	0.0926
BTC	0.0000	0.0000	0.0612	0.0000
ETH	0.0212	0.0000	0.0704	0.0000

A noteworthy aspect of the framework is its adaptability and its forward-looking nature. By automatically detecting the current regime based on the latest available data and using the transition probabilities, the tactical allocation adapts continuously, producing an expected portfolio for the next period as a weighted blend of the regime-specific optimal portfolios. This implementation demonstrates how the full process, from regime detection to probability estimation to portfolio construction, operates as an integrated system rather than as isolated tasks.

To make this dynamic allocation approach concrete, the final implementation step uses the current regime detection together with the Markov transition matrix to produce the expected portfolio weights for the next trading day. As per the results of this study, the latest data identifies the market in Regime 4, thus the transition probabilities imply an expected blend of regime-specific portfolios, resulting in forward-looking weights such as 39% allocations to mid-cap equities of IMCV and IJJ, 10.5% to the SP500, and moderate weights to other large-cap or industrial exposures. The resulting next-day portfolio remains well diversified yet sensitive to the most probable near-term regime shifts, with minimal allocations to assets that historically detract from risk-adjusted performance in similar conditions, such as cryptocurrencies in this regime. This final step illustrates the complete link from unsupervised regime detection to tactical portfolio action, providing a clear example of how the framework helps investors move beyond static, backward-looking asset allocations toward an adaptive, probability-weighted investment process.

Table 4 - Expected weights for next day (31/12/2024) starting from Regime 4

Asset	Weight
SP500	0.1052
IMCV	0.3920
IJJ	0.3920
IWM	0.0065
EEM	0.0063
VV	0.0028
IYJ	0.0012
EFV	0.0938
BTC	0.0000
ETH	0.0003

Altogether, these results provide evidence that a quantitative regime-switching approach can generate clear, data-driven guidance for adjusting asset allocation in response to evolving market conditions. The empirical findings illustrate that distinct market regimes can be detected, that regime transitions can be modelled probabilistically, and that these insights can be translated into systematic tactical portfolio adjustments.

The next chapter will reflect more broadly on the significance of these results in relation to the research question, discuss the limitations inherent in the study's design, and suggest directions for future research and real-world applications.

5. CONCLUSIONS AND FUTURE WORKS

This thesis set out to address a central question in quantitative finance: *Can financial market regimes be identified and forecasted in a way that improves portfolio decision-making?* Motivated by the observation that financial markets do not behave uniformly over time but instead oscillate through distinct states, or *regimes*, the study sought to develop a systematic framework that captures these nonlinear dynamics and translates them into actionable allocation decisions.

To tackle the objective, the research combined unsupervised regime detection through Gaussian Mixture Models (GMM), with probabilistic regime forecasting using a Markov transition matrix. This combination allowed the identification of clear, economically interpretable regimes, and the estimation of transition probabilities that reflect the gradual and uncertain nature of market shifts. By integrating these elements into a regime-weighted tactical asset allocation process, the study moves beyond traditional static strategies, providing a structured approach that proactively adjusts portfolio weights in anticipation of evolving market conditions.

Through the application of a Gaussian Mixture Model, the market was partitioned into interpretable regimes, identified using key descriptors such as moving averages, risk-adjusted returns, volatility, volume, skewness, and kurtosis. The resulting regimes demonstrated clear economic significance, ranging from periods of moderate returns with low volatility to turbulent conditions characterized by heightened downside risk and negative Sharpe ratios.

Building on this segmentation, the study employed a Markov transition matrix to model the probabilistic nature of regime shifts. This addition ensured that regime detection was not merely descriptive but actionable: the matrix quantified the likelihood of moving between states, enabling the portfolio to be adjusted dynamically in anticipation of probable market evolutions. In practical terms, this means that the tactical allocation is not fixed but recalibrated daily using the estimated transition probabilities. By combining regime-specific optimal portfolios through a probability-weighted blend, the approach respects both the current market state and the expected trajectory in the near term.

The empirical results demonstrated that the proposed framework can indeed robustly identify distinct regimes with stable statistical properties and translate these insights into differentiated portfolio allocations. In particular, the integration of the transition matrix allows the portfolio to adapt in a forward-looking manner, which is a meaningful improvement over static or purely backward-looking strategies. This output offers practical guidance for investors navigating uncertain markets, seeking to balance risk and return under changing market conditions. The findings on the next-day expected weights underscore the practical contribution of the framework: by systematically linking regime detection, transition forecasting, and portfolio construction, the approach helps bridge the gap between complex market dynamics and real-world portfolio management. In practise, an investor using this

system would monitor the estimated current regime, apply the transition matrix to assess the likelihood of shifting to alternative states, and adjust their portfolio weights accordingly.

While the findings underscore the promise of a regime-switching approach, it is important to acknowledge the limitations inherent in this study. First, the clustering and transition estimation rely heavily on historical data and the assumption that past relationships will hold in the future, an assumption that may not always be valid, especially during structural market breaks or unprecedented shocks (Damásio & Nicolau, 2024). Second, the transition probabilities are based on a first-order Markov process, which assumes that the next state depends only on the current state, potentially oversimplifying complex market dynamics. Moreover, the asset universe studied, while diversified across equities and cryptocurrencies, remains limited in scope; adding broader asset classes such as fixed income, commodities like gold, or real estate, could yield additional insights into optimal allocations across a wider range of economic environments.

From a policy perspective, the framework's key implication is its potential to inform institutional investors, pension funds, or multi-asset managers in designing dynamic risk management processes that are more responsive to market conditions. However, practical implementation must also weigh transaction costs, rebalancing frequency, and operational feasibility. In real-world settings, overly frequent adjustments might erode returns due to costs and slippage, suggesting the need for careful calibration when deploying regime-switching models in practice.

This research opens several avenues for future work. Methodologically, future studies could explore higher-order Markov chains, which consider multiple prior states and may better capture regime persistence and complex path dependencies. Advanced implementations using the *GenMarkov* package (Damásio & Vasconcelos, 2022) could provide more flexible transition structures and potentially improve predictive accuracy. Extending the analysis to incorporate a broader set of assets including alternative investments such as real estate, gold, or infrastructure, would help test whether the observed regime-specific allocation patterns generalize to different asset universes and market contexts. Additionally, future research could investigate how the framework performs when integrated with other adaptive techniques, such as time-varying covariances or machine learning-based regime detection methods.

In conclusion, the findings lend empirical support to the central research question by showing that it is indeed feasible to identify and forecast market regimes in a manner that directly informs portfolio construction. By explicitly acknowledging that market conditions evolve nonlinearly and probabilistically, the regime-switching tactical asset allocation framework developed in this study illustrates a clear link between complex market dynamics and practical, implementable investment decisions. While the results do not contradict the established literature, they provide an important contribution by demonstrating how quantitative regime models can be operationalized within a coherent risk-return optimization

process, ultimately helping investors better navigate the uncertainties inherent in financial markets.

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